Theme and chassis layout

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Make a dimensioned layout that supports the circuit and drivetrain.

SUCCESS CRITERIA - I can...

1. Label component locations and clearances.
2. Use dimensions to check that parts fit.
3. Explain a structural choice with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A motor uses 4 W for 6 s. Find electrical energy and name a possible non-mechanical output.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: available length = chassis length - components - clearances.
Use measured dimensions for a top view. Keep axles parallel and controls accessible. Get approval before tools/cuts.

Worked reasoning: A 20 cm chassis reserves 6 cm for the battery, 5 cm for the motor and 2 cm total for end clearances. The remaining 7 cm must cover other components and gaps.

Your turn: Draw a dimensioned top view showing these regions. Label any assumed dimensions rather than pretending they were measured.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a dimensioned top view of your chassis. Label motor, battery, switch, axle lines and wheel clearance. Mark the intended forward direction.



Marzano check - circle: ① ② ③

04 Apply the physics

A sample chassis is 18 cm long. A battery needs 6 cm and the motor region needs 5 cm along its length. Mark those regions and calculate the space left before gaps. On your actual plan, add measured clearances and distinguish them from these sample dimensions.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Your theme adds a heavy decoration far to one side. Predict a possible effect and propose a way to test it without claiming the effect is certain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A total current of 16 A splits between two identical parallel branches. Find each branch current.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Name the measurement you will verify before the first cut and record the teacher checkpoint.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Fit the same components on an 18 cm chassis. Propose two layouts and identify a measurement needed before cutting.



Marzano check - circle: 1 2 3